Differentiation Rules: Power, Product, Quotient, Chain

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Power Rule and Basic Operations

Evaluate $\int_2^5 \frac{2x+1}{(x-1)(x+4)}\,dx$.

Decompose $\frac{2x+1}{(x-1)(x+4)} = \frac{7}{5(x+4)} + \frac{3}{5(x-1)}$, integrate to logarithms, and evaluate: $-\frac{\ln(2)}{5} + \frac{7\ln(3)}{5}$.

2 [Foundational] • Target: Product and Quotient Rules

Evaluate $\int_1^5 \frac{1}{(x+1)(x+4)}\,dx$.

Decompose $\frac{1}{(x+1)(x+4)} = -\frac{1}{3(x+4)} + \frac{1}{3(x+1)}$, integrate to logarithms, and evaluate: $-\frac{\ln(3)}{3} + \frac{\ln(5)}{3}$.

3 [Foundational] • Target: The Chain Rule

For parametric $x = e^t\cos t$, $y = e^t\sin t$, find $\frac{d^2y}{dx^2}$ at $t = 0$.

First, $\frac{dy}{dx} = \frac{\sin t + \cos t}{\cos t - \sin t}$. Then $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{2}{(\cos t - \sin t)^2}$. Dividing by $\frac{dx}{dt} = e^t(\cos t - \sin t)$ at $t = 0$ gives $\frac{d^2y}{dx^2}\big|_{t=0} = 2$.

4 [Foundational] • Target: Trigonometric Derivatives with Chain Rule

If $z = x^3 + 3xy^2$ with $x = t^2$ and $y = \sin(t)$, find $\frac{dz}{dt}$ at $t = 1$.

$\frac{dz}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt}$. At $t = 1$: $3\sin^2(1) + 3\cdot 2 + 6\sin(1)\cdot\cos(1) = -3\cos(2) + 3\sin(2) + 9$.

5 [Foundational] • Target: Power Rule and Basic Operations

If $\sin(\theta) = -\frac{24}{25}$ and θ lies in quadrant III, find $\sin 2\theta$.

In quadrant III, $\cos(\theta) = -\frac{7}{25}$ (from $\sin^2\theta + \cos^2\theta = 1$ with the quadrant's sign). Then $\sin 2\theta = \frac{336}{625}$ using the double-angle formula.

6 [Foundational] • Target: Product and Quotient Rules

If $\sin(\theta) = \frac{5}{13}$ and θ lies in quadrant II, find $\cos 2\theta$.

In quadrant II, $\cos(\theta) = -\frac{12}{13}$ (from $\sin^2\theta + \cos^2\theta = 1$ with the quadrant's sign). Then $\cos 2\theta = \frac{119}{169}$ using the double-angle formula.

7 [Foundational] • Target: The Chain Rule

Let $G(x) = \int_0^{\sin(x)} \cos(t)\,dt$. Find $G'(-1)$.

By FTC Part 1 and the chain rule, $G' = \cos(x)\cos(\sin(x))$. At $x = -1$ this is $\cos(1)\cos(\sin(1))$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Coffee at 70°C is placed in a 20°C room and cools to 60°C in 10 min (Newton's law of cooling). How long (min) until it reaches 40°C?

$T-20=50e^{-kt}$ with $k=\frac{\ln\left(\frac{5}{4}\right)}{10}$. Solve $50e^{-kt}=20$: $t=\frac{10\ln\left(\frac{5}{2}\right)}{\ln\left(\frac{5}{4}\right)}$.

9 [Intermediate] • Target: Power Rule and Basic Operations

Find the sum $\sum_{n=1}^{\infty}\frac{2}{n(n+4)}$.

Partial fractions: $\frac{2}{n(n+4)}=-\frac{1}{2(n+4)}+\frac{1}{2n}$. The series telescopes, leaving the first 4 surviving terms: $\frac{25}{24}$.

10 [Intermediate] • Target: Product and Quotient Rules

Water pours into an inverted cone (radius 3 m, height 9 m) at 3 m³/min. How fast is the depth rising when it is 8 m? (m/min)

By similar triangles $r=\frac{h}{3}$, so $V=\frac{\pi h^3}{27}$. Then $\frac{dV}{dt}=\pi\frac{9}{81}h^2\frac{dh}{dt}$, which gives $\frac{27}{64\pi}$ m/min at $h=8$.

11 [Intermediate] • Target: The Chain Rule

Product rule for $f=x^2e^x$:.

Final Answer: $e^x(x^2+2x)$. Applying Derivatives principles, $2xe^x+x^2e^x=e^x(x^2+2x)$. Therefore, the verified correct option is $e^x(x^2+2x)$.

12 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Chain rule: $(\sin(e^x))'=$.

Final Answer: $e^x\cos(e^x)$. Applying Derivatives principles, $\cos(e^x)\cdot e^x$. Therefore, the verified correct option is $e^x\cos(e^x)$.

13 [Intermediate] • Target: Power Rule and Basic Operations

Chain rule: $w=f(x,y,z)$, $x=e^s$, $y=e^s\sin t$, $z=e^s\cos t$. $\partial w/\partial s|_{s=0,\,t=0}$:.

Final Answer: $f_x+f_y\sin t+f_z\cos t$ evaluated at $s=0,t=0$. $w_s=f_xe^s+f_ye^s\sin t+f_ze^s\cos t$; at $s=0,t=0$: $(x,y,z)=(1,0,1)$.

14 [Intermediate] • Target: Product and Quotient Rules

$\frac{d}{dx}[f(g(x))]$ equals (chain rule):.

Final Answer: $f'(g(x))\cdot g'(x)$. Chain rule: outer derivative evaluated at inner × inner derivative.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Chain Rule • Cross-Lesson: 12.01 Convergence Tests

Quotient Rule: $\frac{d}{dx}\left(\frac{x}{x+1}\right) =$.

Final Answer: $\frac{1}{(x+1)^2}$. $\frac{1\cdot(x+1)-x\cdot 1}{(x+1)^2} = \frac{1}{(x+1)^2}$.

16 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 12.02 Power Series and Taylor Series

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$.

Final Answer: all are equivalent. All three expressions are equivalent: $9x^2+2x-6$.

17 [Challenging] • Target: Power Rule and Basic Operations • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

$\frac{d}{dx}(x^2-3)^4 =$ (chain rule).

Final Answer: $8x(x^2-3)^3$. Chain rule: $4(x^2-3)^3\cdot 2x = 8x(x^2-3)^3$.

18 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Product Rule: $\frac{d}{dx}(x\sin x) =$.

Final Answer: $\sin x + x\cos x$. $1\cdot\sin x + x\cdot\cos x = \sin x + x\cos x$.

19 [Challenging] • Target: The Chain Rule • Cross-Lesson: 15.01 Double Integrals

Chain Rule: $\frac{d}{dx}\sin(x^2) =$.

Final Answer: $2x\cos(x^2)$. Chain rule: $\cos(x^2)\cdot 2x = 2x\cos(x^2)$.

20 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Product of two power series $A(x)=\sum a_n x^n$ and $B(x)=\sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:.

Final Answer: $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product). Applying Sequences and Series principles, Cauchy product convolution. Therefore, the verified correct option is $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Power Rule and Basic Operations

Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$ ✓

b. $20x^4 - 9x^2$

c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

By the power rule: $\frac{d}{dx}(4x^5) = 20x^4$, $\frac{d}{dx}(-3x^3) = -9x^2$, $\frac{d}{dx}(7x) = 7$, $\frac{d}{dx}(-9) = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Power Rule and Basic Operations

Differentiate $g(x) = \frac{4}{\sqrt{x}} + \frac{3}{x^2}$.

a. $-2x^{-3/2} - 6x^{-3}$ ✓

b. $-2x^{-1/2} - 6x^{-3}$

c. $2x^{-3/2} - 6x^{-3}$

d. $-4x^{-3/2} - 3x^{-3}$

Rewrite as $4x^{-1/2} + 3x^{-2}$. $g'(x) = 4(-1/2)x^{-3/2} + 3(-2)x^{-3} = -2x^{-3/2} - 6x^{-3}$.

23 [Foundational] • Target: Power Rule and Basic Operations

Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.

a. $x = -1$ **and** $x = 3$ ✓

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

$y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$.

24 [Foundational] • Target: Product and Quotient Rules

Differentiate $h(x) = (x^2 + 3)(2x - 5)$.

a. $6x^2 - 10x + 6$ ✓

b. $6x^2 - 10x$

c. $4x^2 - 10x + 6$

d. $6x^2 + 6$

Product rule: $h'(x) = (2x)(2x - 5) + (x^2 + 3)(2) = 4x^2 - 10x + 2x^2 + 6 = 6x^2 - 10x + 6$.

25 [Foundational] • Target: Product and Quotient Rules

Find $\frac{dy}{dx}$ for $y = \frac{3x - 1}{2x + 5}$.

a. $\frac{17}{(2x + 5)^2}$ ✓

b. $\frac{1}{(2x + 5)^2}$

c. $\frac{6x + 17}{(2x + 5)^2}$

d. $\frac{3}{2}$

Quotient rule: $\frac{3(2x + 5) - (3x - 1)(2)}{(2x + 5)^2} = \frac{6x + 15 - 6x + 2}{(2x + 5)^2} = \frac{17}{(2x + 5)^2}$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Product and Quotient Rules

If $f(2)=3, f'(2)=-1, g(2)=4, g'(2)=2$, find $\left(\frac{f}{g}\right)'(2)$.

a. $-10/16=-5/8$ ✓

b. $2/16=1/8$

c. $-2/16$

d. $-1/2$

$\frac{f'(2)g(2)-f(2)g'(2)}{[g(2)]^2}=\frac{(-1)(4)-(3)(2)}{4^2}=\frac{-4-6}{16}=-\frac{10}{16}=-\frac{5}{8}$.

27 [Foundational] • Target: The Chain Rule

Find $\frac{d}{dx}[(3x^2-5x+2)^4]$.

a. $4(3x^2-5x+2)^3(6x-5)$ ✓

b. $4(3x^2-5x+2)^3$

c. $12(6x-5)^3$

d. $4(6x-5)^3$

By chain rule: $\frac{d}{du}[u^4]\cdot u'=4u^3\cdot(6x-5)=4(3x^2-5x+2)^3(6x-5)$.

28 [Foundational] • Target: The Chain Rule

Find $\frac{dy}{dx}$ for $y=\sqrt{4-x^2}$.

a. $\frac{-x}{\sqrt{4-x^2}}$ ✓

b. $\frac{x}{\sqrt{4-x^2}}$

c. $\frac{-1}{2\sqrt{4-x^2}}$

d. $\frac{-2x}{\sqrt{4-x^2}}$

$y=(4-x^2)^{1/2}\Rightarrow y'=\frac{1}{2}(4-x^2)^{-1/2}(-2x)=\frac{-x}{\sqrt{4-x^2}}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Chain Rule

If $h(x)=f(g(x))$ with $g(3)=5, g'(3)=4$, and $f'(5)=-2$, find $h'(3)$.

a. -8 ✓

b. 8

c. -10

d. 20

$h'(x)=f'(g(x))g'(x)\Rightarrow h'(3)=f'(g(3))g'(3)=f'(5)(4)=(-2)(4)=-8$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Differentiate $y = \sin(3x^2)$.

a. $6x\cos(3x^2)$ ✓

b. $3\cos(3x^2)$

c. $\cos(6x)$

d. $6x\sin(3x^2)$

Chain rule: $\frac{d}{dx}[\sin(u)] = \cos(u) \cdot u' = \cos(3x^2) \cdot 6x = 6x\cos(3x^2)$.

31 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Find $\frac{d}{dx}[\tan^3(x)]$.

a. $3\tan^2(x)\sec^2(x)$ ✓

b. $3\tan^2(x)$

c. $3\sec^4(x)$

d. $\sec^6(x)$

Let $u = \tan x$: $\frac{d}{dx}[u^3] = 3u^2u' = 3\tan^2(x)\sec^2(x)$.

32 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$ ✓

b. $4\cos(2x)$

c. $-2\sin(2x)$

d. $-4\sin(2x)$

$y' = -2\sin(2x)$. $y'' = \frac{d}{dx}[-2\sin(2x)] = -2(2\cos(2x)) = -4\cos(2x)$.

33 [Intermediate] • Target: Power Rule and Basic Operations

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2 ✓

c. 0

d. 1

Differentiating $x^2 + y^2 = 2$ gives $y' = -\frac{x}{y}$ and $y'' = -\frac{y - xy'}{y^2} = -\frac{x^2 + y^2}{y^3}$. At $(1, 1)$, $y'' = -\frac{2}{1^3} = -2$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Product and Quotient Rules

Cauchy product: $(\sum_{n=0}^{\infty} x^n)^2 = \sum_{n=0}^{\infty} (n+1)x^n$. This equals:

a. $1/(1-x)^2$ ✓

b. $1/(1-x)$

c. $(1-x)^2$

d. $2/(1-x)$

Applying Convergence Tests and Power Series principles, $(1/(1-x))^2 = \sum (n+1)x^n$. Therefore, the verified correct option is $1/(1-x)^2$.

35 [Intermediate] • Target: The Chain Rule

Implicit differentiation of $F(x, y) = 0$: $\frac{dy}{dx} =$

a. $-F_x/F_y$ ✓

b. F_x/F_y

c. $-F_y/F_x$

d. F_{xy}

Applying Applications of Derivatives principles, $dy/dx = -F_x/F_y$. Therefore, the verified correct option is $-F_x/F_y$.

36 [Intermediate] • Target: Trigonometric Derivatives with Chain Rule

Implicit differentiation of $x^2 + y^2 = 25$: $dy/dx =$

a. $-x/y$ ✓

b. x/y

c. $-y/x$

d. $2x$

Applying Derivatives principles, $2x + 2yy' = 0 \Rightarrow y' = -x/y$. Therefore, the verified correct option is $-x/y$.

37 [Intermediate] • Target: Power Rule and Basic Operations

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum nx^{n-1}/(n+1)$

c. $1/(1-x)$ ✓

d. $\ln(1-x)$

Recognise $\int \frac{1}{1-x} dx = -\ln(1-x) = \sum x^n/(n+1)$; differentiate $\to 1/(1-x)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Logarithmic differentiation: if $y = \frac{x^2\sqrt{x+1}}{e^x}$, then $y' =$

a. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} - 1\right)$ ✓

b. $y\left(\frac{2}{x} - \frac{1}{2(x+1)} - 1\right)$

c. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} + 1\right)$

d. $y\left(2x + \frac{1}{2} - 1\right)$

$\ln y = 2\ln x + \frac{1}{2}\ln(x+1) - x$. Differentiate: $y'/y = 2/x + 1/(2(x+1)) - 1$.

39 [Challenging] • Target: The Chain Rule • Cross-Lesson: 15.01 Double Integrals

Logarithmic differentiation: $y = x^x$. $y' =$

a. $x^x(1 + \ln x)$ ✓

b. $x \cdot x^{x-1}$

c. $x^x \ln x$

d. e^x

$\ln y = x\ln x \Rightarrow y'/y = 1 + \ln x \Rightarrow y' = x^x(1 + \ln x)$.

40 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

b. are zero

c. **exist** ✓

d. diverge

Power series define smooth (C^∞) functions inside R.

41 [Challenging] • Target: Power Rule and Basic Operations • Cross-Lesson: 2.01 Evaluating Limits

Differentiation of geometric series $1/(1-x) = \sum x^n$ gives $1/(1-x)^2 =$

a. $\sum n x^{n-1}$

b. $\sum (n+1)x^n$ ✓

c. $\sum x^{n-1}$

d. $\sum n x^n$

$d/dx \sum x^n = \sum n x^{n-1} = \sum (n+1)x^n$ (shift n).

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 2.02 Continuity

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

b. $\frac{y}{x}$

c. $-\frac{y}{x}$ ✓

d. $\frac{x}{y}$

$y + x\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -y/x$.

43 [Challenging] • Target: The Chain Rule • Cross-Lesson: 3.01 The Derivative from First Principles

Leibniz rule: $(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$. Analogy with:

a. **binomial theorem** ✓

b. FTC

c. MVT

d. Taylor series

Leibniz general product rule ↔ binomial theorem.

44 [Challenging] • Target: Trigonometric Derivatives with Chain Rule • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Term-by-term differentiation of $\sum a_n x^n$ within $|x| < R$ gives:

a. $\sum n a_n x^n$

b. $\sum a_{n+1} x^{n+1}$

c. $\sum a_n x^{n+1}/(n+1)$

d. $\sum n a_n x^{n-1}$ ✓

Differentiate term-by-term: $\sum n a_n x^{n-1}$.

45 [Challenging] • Target: Power Rule and Basic Operations • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

b. 1/0

c. 0/0 ✓

d. 1/1

Indeterminate form $0/0$ or ∞/∞.

© ClipSAT — clipsat.org | All Rights Reserved